

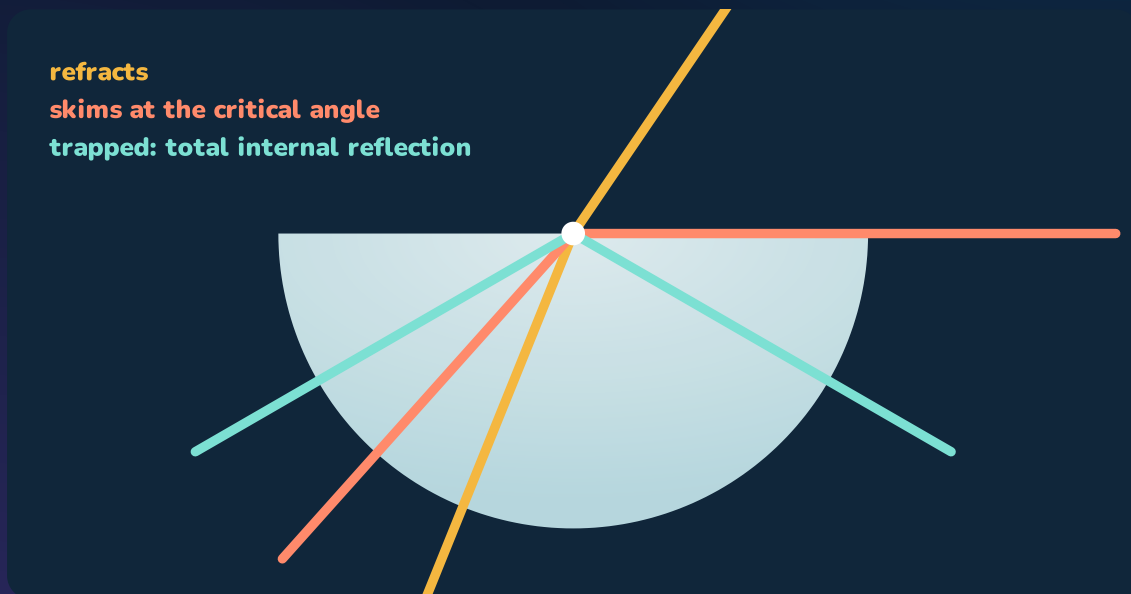
Tutorial 5

Light

Reflection · Refraction · Total internal reflection · Thin converging lenses

"All these fifty years of conscious brooding have brought me no nearer to the answer to the question, 'What are light quanta?'"

Albert Einstein, letter to Michele Besso, 1951



Infographic placeholder · drop the NotebookLM infographic for this chapter here

Paper 1 · 20 marks · 30 min

Paper 2 · 40 marks · 45 min

Total · 60 marks

Concept summary

Reflection

- Angles are measured from the **normal** (at 90° to the surface).
- Law: **$i = r$** ; incident ray, reflected ray and normal lie in one plane.
- Plane mirror image: **virtual, upright, same size, laterally inverted**, as far behind the mirror as the object is in front.

Refraction

- Bending of light when it changes **speed** passing from one medium to another.
- Into a denser medium (slower): bends **towards** the normal. Out of it: **away**.
- Along the normal: no bending, but speed still changes.

Refractive index

- **$n = \sin i / \sin r$** (i in air, r in the medium). Snell's law: the ratio is constant.
- **$n = c / v$** , $c = 3.0 \times 10^8$ m/s. n has no unit and is always > 1 .
- Glass 1.5, water 1.33, diamond 2.42.

Total internal reflection

- Needs **dense** $\rightarrow$ **less dense** AND **$i > c$** .
- **$\sin c = 1 / n$** . Glass $c = 41.8^\circ$, water $c = 48.8^\circ$.
- Optical fibres: core has higher n than cladding. Uses: telecom, endoscopes. More data, less loss, no electrical interference, secure.

Thin converging lens

- **Focal length f** : optical centre to principal focus F .
- Rays: parallel $\rightarrow$ through F ; through C $\rightarrow$ straight; through near F $\rightarrow$ parallel.
- Beyond $2F$: real, inverted, diminished (camera). At $2F$: same size. F – $2F$: magnified (projector). Inside F : virtual, upright, magnified (magnifying glass).

Exam pitfalls

- Angle given from the **surface**? Use 90° – angle.
- $n < 1$ or a Math ERROR means a fraction is upside down.
- Ray diagrams: ruler, arrows, virtual lines **dashed**, image labelled.
- Real/virtual, upright/inverted, magnified/diminished: give all three.

Tutorial 5 · Light

Paper 1 · Multiple Choice

30 minutes · 20 marks

Name	
Class	
Register number	

READ THESE INSTRUCTIONS FIRST

- Write your name, class and register number in the spaces above.
- There are **twenty** questions in this paper. Answer **all** questions.
- For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the one you consider correct and write its letter in the brackets provided.
- Each correct answer scores one mark. A mark is not deducted for a wrong answer.
- Any rough working should be done in this booklet. The use of an approved scientific calculator is expected, where appropriate.
- Take the speed of light in a vacuum as 3.0×10^8 m/s.

- 7** A ray of light in air strikes a block at an angle of incidence of 45° . The angle of refraction is 28° . What is the refractive index of the block?

A 0.66 **B** 1.33
C 1.51 **D** 1.61 ()

8 A ray of light passes from water into air. The angle of incidence is less than the critical angle. What happens to the ray at the surface?

A It bends towards the normal and speeds up.
B It bends towards the normal and slows down.
C It bends away from the normal and slows down.
D It bends away from the normal and speeds up. ()

9 A material has a refractive index of 1.25. What is its critical angle?

A 53° **B** 37° **C** 39° **D** 51° ()

10 Which conditions are needed for total internal reflection?

 - I Light travels from an optically denser medium towards a less dense medium.
 - II The angle of incidence is greater than the critical angle.
 - III The light must be of one colour only.

A I only **B** I and II only
C II and III only **D** I, II and III ()

11 Why is the cladding of an optical fibre made of glass with a lower refractive index than the core?

A So that it absorbs any light escaping from the core.
B So that the fibre is flexible enough to bend.
C So that light striking the core–cladding boundary at large angles is totally internally reflected.
D So that light refracts into the cladding and travels faster. ()

12 The critical angle of a glass is 42° . A ray inside the glass strikes the glass–air surface at an angle of incidence of 50° . What happens to the ray?

A It refracts out, bending towards the normal. **B** It travels along the surface.
C It passes straight through. **D** It is totally internally reflected. ()

13 What is an advantage of using optical fibres rather than copper cables for telecommunication?

- A** They carry much more information with less signal loss.
- B** They are easier to repair when broken.
- C** They can carry electric current as well as light.
- D** They can be bent at sharp right angles without losing any light.

()

14 The focal length of a converging lens is the distance between

- A** the lens and the image.
- B** the optical centre and the principal focus.
- C** the object and the image.
- D** the principal focus and $2F$.

()

15 Which row correctly describes the image formed by a thin converging lens?

	object position	nature of image
A	beyond $2F$	virtual, upright, diminished
B	at $2F$	real, inverted, magnified
C	between F and $2F$	real, inverted, magnified
D	between F and the lens	real, upright, magnified

()

16 An object is placed 10 cm from a converging lens of focal length 10 cm. Where is the image formed?

- A** 10 cm
- B** 20 cm
- C** 5 cm
- D** at infinity

()

17 A converging lens is used as a magnifying glass. What type of image does it form?

- A** virtual, upright and magnified
- B** real, inverted and magnified
- C** virtual, inverted and diminished
- D** real, upright and the same size

()

18 Which statements about a real image are correct?

- I** It can be formed on a screen.
- II** It is formed where rays of light actually meet.
- III** It is always upright.

- A** I only
- B** I and II only
- C** II and III only
- D** I, II and III

()

19 An object is placed 20 cm from a converging lens of focal length 10 cm. How far from the lens is the image?

A 10 cm

B 15 cm

C 20 cm

D 40 cm

()

20 In a slide projector, where is the slide placed relative to the converging lens?

A beyond $2F$

B at $2F$

C at F

D between F and $2F$

()

END OF PAPER 1

Tutorial 5 · Light

Paper 2 · Structured Questions

45 minutes · 40 marks

Name	
Class	
Register number	

READ THESE INSTRUCTIONS FIRST

- Write your name, class and register number in the spaces above.
- **Section A:** answer **all** questions in the spaces provided.
- **Section B:** answer **one** question in the spaces provided.
- Show your working. Omission of essential working will result in loss of marks.
- The number of marks is given in brackets [] at the end of each question or part question.
- Take the speed of light in a vacuum as 3.0×10^8 m/s.

For examiner's use

Section A · Q1		5
Section A · Q2		7
Section A · Q3		6
Section A · Q4		8
Section A · Q5		6
Section B · Q6 / Q7		8
Total		40

Section A Answer **all** questions.

1 This question is about reflection and refractive index.

(a) State the two laws of reflection. **[2]**

.....

.....

.....

(b) Define the refractive index of a medium in terms of the speed of light. **[1]**

.....

.....

(c) Light travels at 2.25×10^8 m/s in water. Calculate the refractive index of water. **[2]**

refractive index =

[Total: 5]

2 A ray of light in air strikes the surface of a rectangular glass block at an angle of incidence of 40° . The refractive index of the glass is 1.52.

(a) Calculate the angle of refraction. **[2]**

angle of refraction = $^\circ$

(b) Calculate the speed of light in the glass. **[2]**

speed = m/s

(c) The ray leaves through the opposite, parallel face of the block. State and explain the direction of the emergent ray compared with the incident ray. **[2]**

.....

.....

.....

- (d) Explain why a ray that strikes the block along the normal does not bend. [1]

.....

.....

[Total: 7]

- 3 An endoscope uses optical fibres made of glass of refractive index 1.48.

- (a) Define critical angle. [1]

.....

.....

- (b) Calculate the critical angle of the glass. [2]

critical angle =°

- (c) Explain how light travels along a curved optical fibre. [2]

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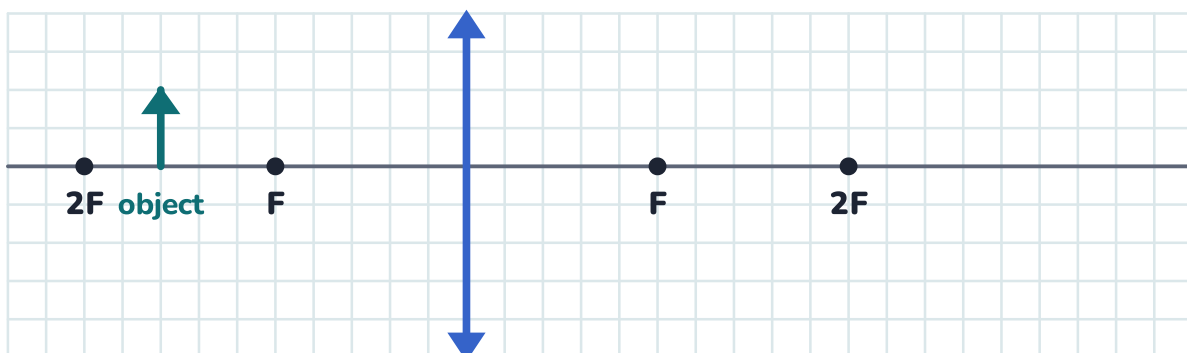
- (d) State one advantage of using an endoscope instead of surgery to examine a patient's stomach. [1]

.....

.....

[Total: 6]

- 4 An object 2.0 cm tall is placed 8.0 cm from a thin converging lens of focal length 5.0 cm. The grid is drawn to scale: 1 square represents 1 cm.



Scale: 1 square = 1 cm

- (a) Complete the ray diagram to locate the image. Label the image. [3]
- (b) Measure the distance of the image from the lens. [1]

image distance = cm

- (c) Measure the height of the image. [1]

height = cm

- (d) State three characteristics of the image. [2]

.....

- (e) Name one optical device that uses this arrangement. [1]

.....

[Total: 8]

- 5 A small lamp is fixed to the bottom of a swimming pool. The refractive index of water is 1.33.

- (a) Calculate the critical angle for water. [2]

critical angle = °

- (b) A ray from the lamp strikes the water surface at an angle of incidence of 30°. [2]
 Calculate the angle the ray makes with the normal in the air.

angle = °

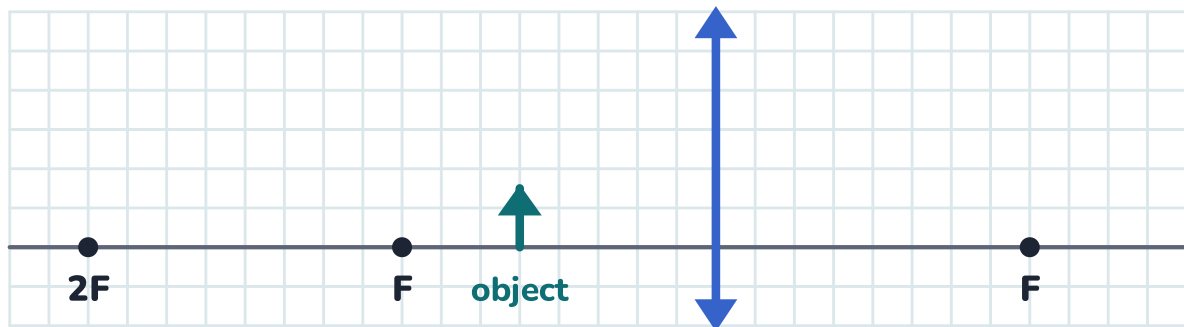
- (c) Another ray strikes the surface at an angle of incidence of 60° . Describe and explain what happens to this ray. [2]

[Total: 6]

Section B

Answer **one** question.

- 6** A student uses a converging lens of focal length 8.0 cm as a magnifying glass to look at a stamp.



Scale: 1 square = 1 cm

- (a)** Define principal focus. **[1]**

.....

.....

- (b)** The stamp detail, 1.5 cm tall, is 5.0 cm from the lens. On the grid (1 square = 1 cm), complete a ray diagram to find the image. Label the image. **[3]**

- (c)** State the distance of the image from the lens and its height. **[2]**

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.....

- (d)** The student moves the lens so that the stamp is 12 cm from it. Describe the image now formed. **[2]**

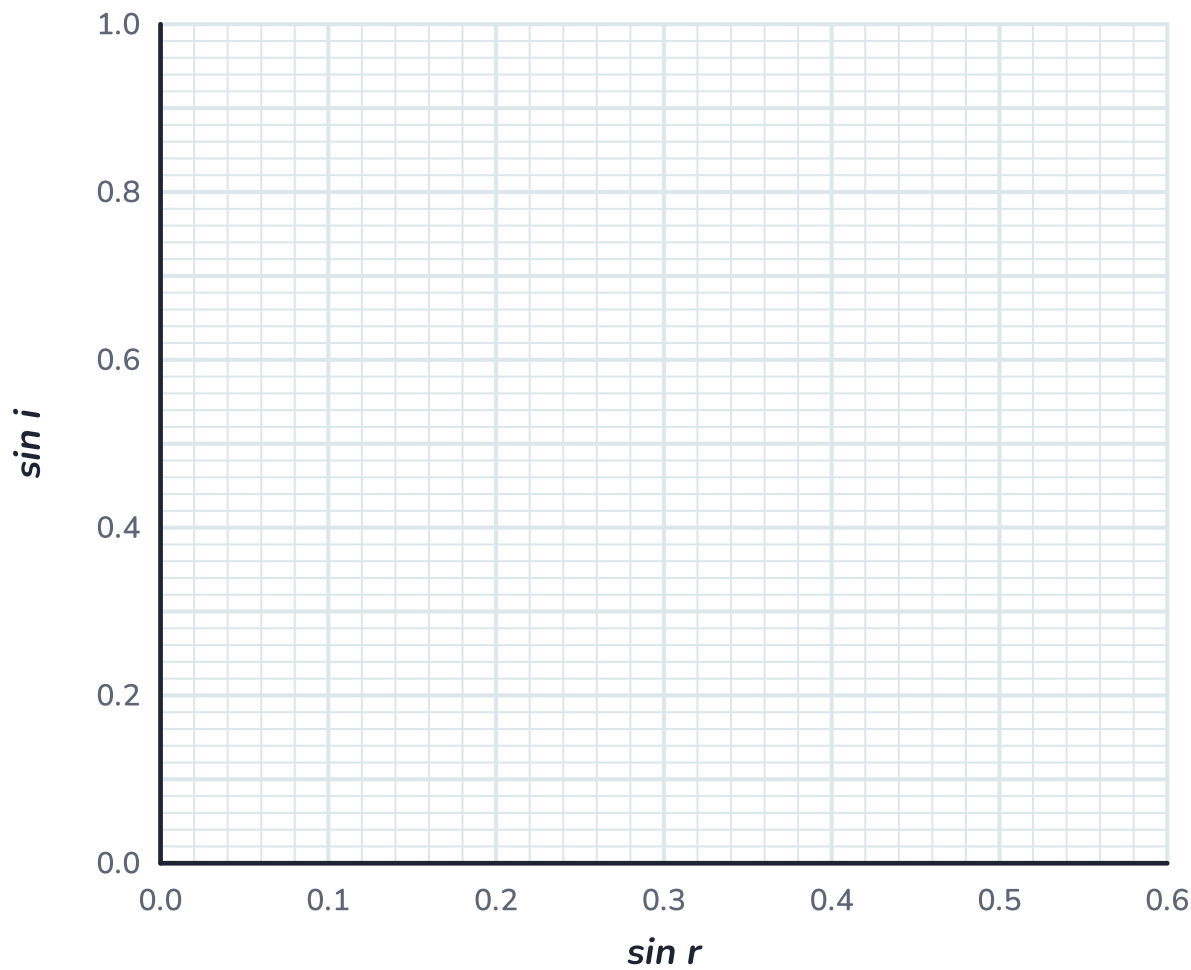
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[Total: 8]

- 7** A student investigates refraction using a ray box and a rectangular glass block. For five angles of incidence i , she measures the angle of refraction r .

$i / ^\circ$	$r / ^\circ$	$\sin i$	$\sin r$
20	13.2	0.342	0.228
30	19.5	0.500	0.334
40	25.4		
50	30.7	0.766	0.511
60	35.3		



- (a)

Describe how she can mark the path of the ray inside the block to measure r .

[2]
- (b)

Complete the table.

[2]

- (c) Plot $\sin i$ against $\sin r$ on the grid and draw the best-fit line. Use it to find the refractive index of the glass. **[3]**

refractive index =

- (d) State one precaution to improve the accuracy of her measurements. **[1]**

.....

.....

[Total: 8]

END OF PAPER 2

Answer key

Paper 1 · Multiple choice

Q	Ans	Why
1	A	Angles are measured from the normal: $90^\circ - 30^\circ = 60^\circ$.
2	B	$i + r = 35^\circ + 35^\circ = 70^\circ$.
3	C	A plane mirror image cannot be caught on a screen, is upright, same size and left-right reversed.
4	D	III is false: light slows down in glass.
5	A	$v = c / n = 3.0 \times 10^8 / 1.6 = 1.9 \times 10^8 \text{ m/s}$.
6	B	Same i , so the smallest r means the greatest bending and the largest n .
7	C	$n = \sin 45^\circ / \sin 28^\circ = 0.707 / 0.469 = 1.51$.
8	D	Light speeds up entering air (less dense), so it bends away from the normal.
9	A	$\sin c = 1 / 1.25 = 0.80$, so $c = 53^\circ$.
10	B	Colour is irrelevant to whether TIR occurs.
11	C	TIR can only happen going from higher n (core) to lower n (cladding).
12	D	$50^\circ > 42^\circ$, so total internal reflection occurs.
13	A	High data capacity and low loss are the key advantages.
14	B	By definition.
15	C	At $2F$ the image is the same size; inside F it is virtual; beyond $2F$ it is real and diminished.
16	D	An object at F gives parallel rays: no image is formed (at infinity).
17	A	Object inside F : virtual, upright, magnified.
18	B	Real images from a single converging lens are inverted.
19	C	The object is at $2F$, so the image is at $2F$: 20 cm.
20	D	Between F and $2F$ gives a real, inverted, magnified image on the screen.

Paper 2 · Question 1 [5]

- (a) The angle of incidence equals the angle of reflection. The incident ray, the reflected ray and the normal lie in the same plane. [2]
- (b) Refractive index = speed of light in a vacuum $\div$ speed of light in the medium. [1]
- (c) 1 Use $n = c / v$. [2]

$$n = 3.0 \times 10^8 / 2.25 \times 10^8 = 1.33$$

Paper 2 · Question 2 [7]

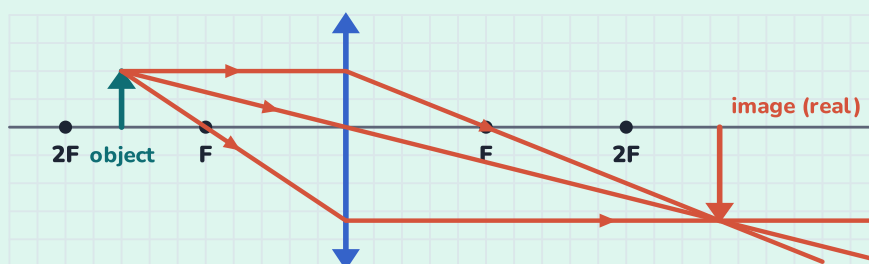
- (a) **1** Rearrange $n = \sin i / \sin r$. [2]
 $\sin r = \sin 40^\circ / 1.52 = 0.643 / 1.52 = 0.423$
- 2** Inverse sine.
 $r = 25.0^\circ$
-
- (b) **1** Use $v = c / n$. [2]
 $v = 3.0 \times 10^8 / 1.52 = 1.97 \times 10^8 \text{ m/s}$
-
- (c) It is **parallel** to the incident ray (displaced sideways). The ray meets the second face at 25.0° , so by reversibility it leaves at 40° to the normal. [2]
-
- (d) Its angle of incidence is 0° , so $\sin r = 0$ and $r = 0^\circ$: it slows down but does not change direction. [1]

Paper 2 · Question 3 [6]

- (a) The angle of incidence in the optically denser medium for which the angle of refraction is 90° . [1]
-
- (b) **1** Use $\sin c = 1 / n$. [2]
 $\sin c = 1 / 1.48 = 0.676$, so $c = 42.5^\circ$
-
- (c) Light strikes the core–cladding boundary at an angle of incidence **greater than the critical angle**, so it is **totally internally reflected**. This repeats all along the fibre, even round bends. [2]
-
- (d) Only a small opening (or none) is needed, so less pain, lower risk of infection and faster recovery. [1]

Paper 2 · Question 4 [8]

- (a) Ray parallel to the axis refracts through F; ray through C is undeviated; they cross at the image tip. [3]



Scale: 1 square = 1 cm

- (b) 13.3 cm (accept 12.5–14.0 cm) [1]

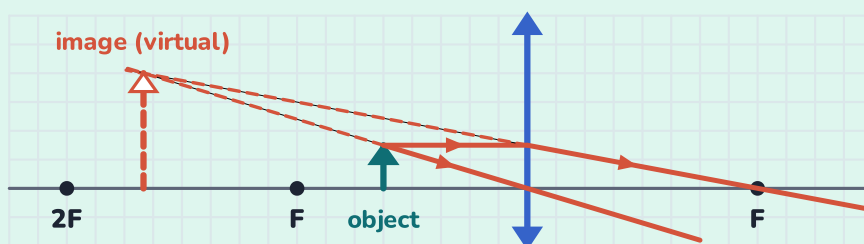
- (c) 3.3 cm (accept 3.0–3.6 cm) [1]
- (d) Real, inverted, magnified. (3 correct: 2 marks; 2 correct: 1 mark) [2]
- (e) A projector. [1]

Paper 2 · Question 5 [6]

- (a) 1 Use $\sin c = 1 / n$. [2]
 $\sin c = 1 / 1.33 = 0.752$, so $c = 48.8^\circ$
- (b) 1 The larger angle is in air. [2]
 $\sin \theta = 1.33 \times \sin 30^\circ = 0.665$
 2 Inverse sine.
 $\theta = 41.7^\circ$
- (c) It is **totally internally reflected** back into the water, because 60° is **greater than the critical angle** of 48.8° and the light is going from water to air. [2]

Paper 2 · Question 6 [8]

- (a) The point on the principal axis where rays parallel to the axis converge after passing through the lens. [1]
- (b) Draw the parallel ray through F and the ray through C; they diverge, so extend both back with dashed lines until they meet. [3]



Scale: 1 square = 1 cm

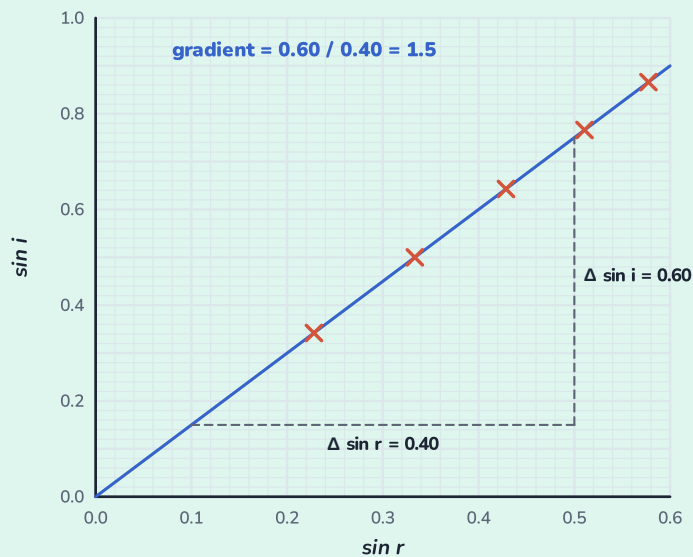
- (c) 13.3 cm from the lens, on the same side as the stamp (accept 12.5–14.0 cm); height 4.0 cm (accept 3.7–4.3 cm). [2]
- (d) The stamp is now between F and 2F, so the image is **real, inverted and magnified**, on the opposite side of the lens (beyond 2F). [2]

Paper 2 · Question 7 [8]

- (a) Mark the incident and emergent rays with crosses (or pins) far apart, remove the block, join the entry and exit points with a straight line, then draw the normal and measure r with a protractor. [2]

- (b) $i = 40^\circ$: $\sin i = 0.643$, $\sin r = 0.429$. $i = 60^\circ$: $\sin i = 0.866$, $\sin r = 0.578$. [2]

- (c) Points lie on a straight line through the origin, so $\sin i / \sin r$ is constant (Snell's law). [3]



- 1 Gradient from a large triangle.
 $n = 0.60 / 0.40 = 1.5$

- (d) Use a narrow ray; read the protractor with the eye directly above; mark the ray at points far apart; repeat and average. [1]